Fiscal Year:	FY 2020	**Task Last Updated:**	FY 11/12/2019
PI Name:	Zea, Luis Ph.D.		
Project Title:	Characterization of Biofilm Formation, Growth, and Gene Expression on Different Materials and Environmental Conditions in Microgravity		
Division Name:	Physical Sciences, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boulder	**State:**	CO
Zip Code:	80309-0429	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	01/03/2017	**End Date:**	06/02/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Strutzenberg Littles, Louise	**Contact Phone:**	(256) 544-0946
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 6/02/2021 per NSSC information (Ed., 9/10/20) NOTE: End date changed to 1/02/2021 per S. Reagan/MSFC (Ed., 10/15/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Klaus, David Ph.D. (University of Colorado, Boulder) Moeller, Ralf Ph.D. (Deutsches Zentrum Fuer Luft- Und Raumfahrt E.V., Germany) Muecklich, Frank Ph.D. (Saarland University, Germany) Stodieck, Louis Ph.D. (University of Colorado, Boulder)		
Grant/Contract No.:	80NSSC17K0036 ; NNX17AC21G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Biofilm formation and its concomitant risk of causing equipment malfunction and human illnesses is a problem that needs to be addressed to enable safe long-term human space exploration. With that objective, this proposal aims to characterize biofilm growth during one International Space Station (ISS) increment in terms of mass, thickness, morphology, and associated gene expression using different spaceflight-relevant microbial species and material substrata. This project also aims to elucidate the biomechanical and transcriptomic mechanisms involved in the formation of the “column-and-canopy” biofilm architecture observed in space. The experiment is designed to assess the expression of genes associated with conferring microorganisms with resistance to oxidative stress, acidity, and antimicrobials in biofilms formed in space, and to determine physical mechanisms of material/microorganisms interaction in biofilms. The experiment will be conducted using BioServe's BioCells housing substrata coupons exposed to different microorganisms and media. Post-flight analyses of fixed samples include confocal laser scanning microscopy and RNA-sequencing. The significance of this project stems from gaining valuable new knowledge that may lead to improved methods and/or material properties for controlling biofilm formation during long duration space missions. The selection of bacterial species, growth media, and substrata material make the results applicable to a) the safety and maintenance of crewed spacecraft, b) planetary protection, c) mitigation of biofilm-associated illnesses on the crew, and d) biofilm-associated illnesses on Earth, thus making it relevant not only to this Research Announcement, but also to NASA’s Human Exploration and Operations Directorate, Space Biology Program, and Human Research Roadmap.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Biofilms are associated with multiple illnesses, including endocarditis (bacterial infection of cardiac tissue), cystitis (a urinary tract infection), and otitis media (an inflammatory disease of the middle ear), as well as several other hospital-acquired infections. Understanding how these biofilms form and their relationship with resistance to antibiotics can help us better prepare novel solutions to these problems on Earth.
Task Progress:	1. All aspects of the experiment design were determined and successfully tested, separately. 2. Experiment design was verified via an experiment verification test (EVT). The results were successful and NASA Headquarters, via a Flight Selection Review, agreed to proceed with flight. 3. A delay in launch from the originally-planned SpaceX-18 to NG-12 required further testing to confirm the samples could survive the longer period before returning to Earth. Results were positive, enabling this payload to proceed with flight on NG-12.
Bibliography Type:	Description: (Last Updated: 11/13/2025)
Abstracts for Journals and Proceedings	Flores P, Schauer R, McBride S, Luo J, Pennington P, Countryman S, Hoehn C, Vellone M, Doraisingam S, Floyd S, Rupert M, Klaus D, Varanasi KK, Muecklich F, Gorti S, Sato K, Moeller R, Stodieck L, Zea L. "Verification of the Design of a Bacterial Biofilm Experiment in Preparation for Spaceflight." 35th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Denver CO, November 20-23, 2019. ASGSR abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Denver CO, November 20-23, 2019. , Nov-2019
Abstracts for Journals and Proceedings	Schauer R, Flores P, Cortesão M, McBride SA, Frank G, Countryman S, Hoehn C, Vellone M, Doraisingam S, Floyd S, Rupert M, Klaus D, Varanasi KK, Moeller R, Sato K, Gorti S, Stodieck L, Zea L. "Verification of the Design of a Fungal Biofilm Experiment in Preparation for Spaceflight." 35th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Denver CO, November 20-23, 2019. ASGSR abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Denver CO, November 20-23, 2019. , Nov-2019